EPFL

Test-Time Compositional Control for Diffusion Models

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Problem

Long Images Generation: *Multi-constraint control at inference time, without retraining*

Left Condition:
Fluffy white rabbit

Right Condition:
lush forest with stream

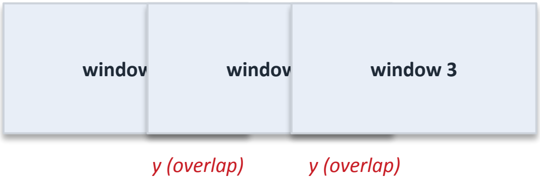
Research question:

How to control the generated result of diffusion models under multiple constraints at inference time?

different left/right semantic prompts and a coherent transition between them

Testbed & Baselines

Testbed: Panoramic Text-Prompt Bridge (SD 1.5)



- **Bridge variable y** = latent overlap between adjacent windows.
- **Endpoints** = left and right text prompts.
- **Controlled comparison** all methods share prompts, seed, schedule, windows — they differ only in how the overlap is composed.

Naive composition

$$\epsilon_{\text{naive}}(u_t, t) = \text{Merge}\left(\epsilon_1, \ldots, \epsilon_N\right)$$

Combining pairwise factors p(x,y)·p(y,z) counts the shared variable y twice, which leads to inconsistency near the seam

Factor graph\ DiffCollage

$$p_{\text{fg}}(x, y, z) \propto \frac{p(x, y)p(y, z)}{p(y)}$$

$$s_{\text{fg}} = s_{xy} \oplus s_{yz} - s_y$$

- Corrects overlap double-counting via marginal subtraction.
- Requires explicit overlap marginal estimation.
- Assumes a tree-style factorization.
- Limited modeling of residual global dependence.

Ours: Bridge Correction & SMC / FKC-PoE

Bridge Correction

1. Start from DiffCollage

$$s_{\text{bridge}} = s_{xy} \oplus s_{yz} - s_{y,\text{implicit}} + \nabla \log R(x, y, z)$$

2. Estimate local $\hat{x}_0$
3. Detect overlap mismatch

$$r_i = \hat{x}_{0,i}^{\text{right}} - \hat{x}_{0,i+1}^{\text{left}}$$

4. Apply symmetric overlap correction

$$\epsilon_{\text{bridge}} = \epsilon_{\text{dc}} + \Delta\epsilon$$

Why it may work

- Fixes the residual bridge mismatch beyond overlap double-counting
- Reduces seam artifacts and encourages smoother local consistency

SMC / FKC-PoE

Sequential Monte Carlo / Feynman–Kac Correctors Product-of-Expert

SMC keeps multiple candidate trajectories, while FKC motivates trajectory reweighting as a correction for biased composition.

1. Start from Naive PoE

$$\epsilon_{\text{PoE}} = \text{Merge}(\epsilon_1, \ldots, \epsilon_N)$$

2. Keep multiple trajectories
3. Reweight them by overlap compatibility

$$G_t^k = \frac{1}{D} \sum_i \langle \epsilon_{i,R}^k(t),\, \epsilon_{i+1,L}^k(t) \rangle$$

$$\log w_t^k = \log w_{t+1}^k + \beta G_t^k$$

4. Resample better particles

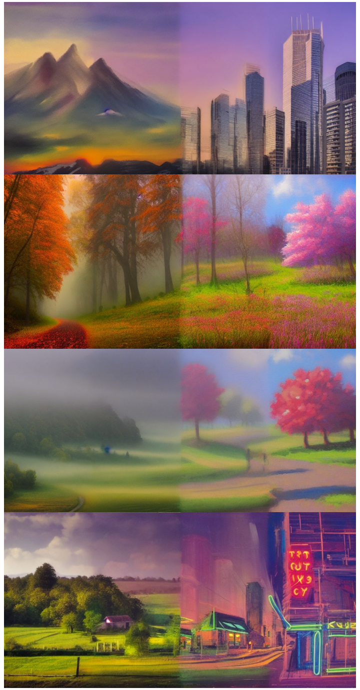
$$k^* = \arg\max_k \log w_0^k, \qquad z_0^* = z_0^{k^*}$$

Why it may work

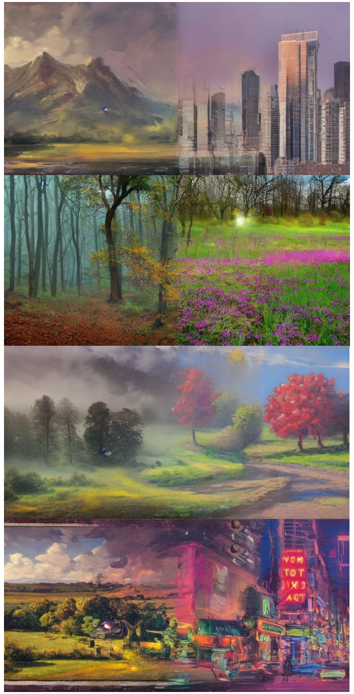
- Addresses the biased single-trajectory PoE composition by reweighting and resampling trajectories according to overlap compatibility
- Goes beyond direct local fixing and favors more globally consistent bridges

Results

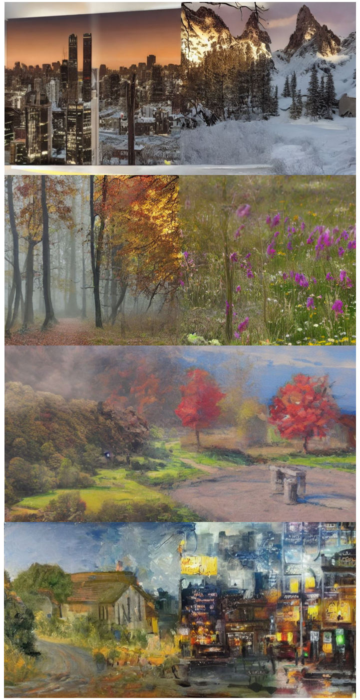
Naive



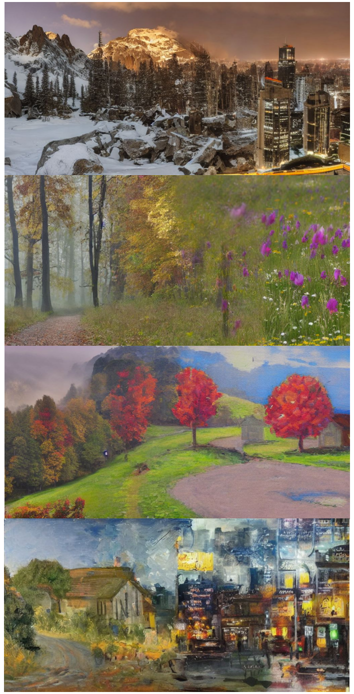
Diffcollage



Bridge Correction(ours)



SMC (ours)



Results

SMC (3 prompts)

Bridge Correction (3 prompts)

Results

Method	Config	Seam MSE mean	Seam MSE max	Seam MSE std
Naive	seed = 42	0.0187	0.0236	0.0163
DiffCollage	seed = 42	0.0175	0.0208	0.0158
Ours: Bridge Correction ($c = 0.01$)	seed = 42	0.0184	0.0223	0.0165
Ours: Bridge Correction ($c = 0.05$, ablation)	seed = 42	0.0628	0.0996	0.0259
Ours: SMC / FKC-PoE ($K = 8$, $\beta = 1.0$)	seed = 17, overlap = 48	0.0198	0.0237	0.0128
Ours: SMC / FKC-PoE ($K = 8$, $\beta = 0.1$, ablation)	seed = 17, overlap = 48	0.0211	0.0257	0.0127

What did we learn from the results?

1. Reproducing baselines closed the gap.

DiffCollage (tuned): 0.0175 vs paper's 0.0307 (−40%).

2. SMC's real win is variance, not mean.

Per-prompt std: SMC 0.0128 vs baselines 0.016 (−22%).

3. The SMC gain came from overlap, not reweighting.

K, β changes < 0.003; overlap 32 → 48 alone gives −35%. β ≥ 10 makes it worse.

4. Bridge Correction has a narrow sweet spot for c.

c = 0.01 → 0.0184; c = 0.05 → 0.0628 (×3.4 collapse).

Conclusion & takeaways

Contributions

• Reproduced DiffCollage / Naive / Bridge Correction baselines to paper-grade.

• New SMC / FKC-PoE method for SD 1.5 bridge composition; 50+ configs swept.

Takeaway

• SMC's real win is –22% per-prompt variance (std 0.0128 vs 0.016), not lower mean. Reproducing baselines properly closes most of the headroom new methods claim.

Limitations & future work

• Seam MSE only; ε-agreement FK reward saturates ($\beta \geq 10$ worse). Need a stronger semantic-space reward.

• SD 1.5 only; SMC tuned on different seed / overlap than baselines.

• Open question: can SMC's variance-reduction transfer to deterministic PoE?